

# LOT SYSTEMS CORPORATION

## LOT COMPUTER

HARDWARE MANUAL & BUILD PLAN

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S-2: VADIK MARMELODOV, Inventor, COSM0® CIA | CLASS: RESTRICTED // S-2 EYES | ISSUE: 2026.07.18  
brand.lot-systems.com | Made in the USA | LOT® Founded 7 April 2016 | COSM0® Founded 1 July 2024

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LOT SYSTEMS CORPORATION  
LOT COMPUTER – HARDWARE SPECIFICATION & BUILD PLAN

DOCUMENT LOT-COMPUTER-HARDWARE-SPEC

ISSUE DATE 2026.07.18

CLASS RESTRICTED // S-2 EYES

STYLE TERMINAL GRID

S-2 VADIK MARMELODOV, Inventor, COSMO® CIA

STATUS READY – v1.0, first surviving iteration (prior brave-lamport COSMO Hardware work, LOT-MANIFEST entry brave-lamport-t9z5u8, +2610 lines / 7 files, was never merged and no longer exists on any remote branch – this document restarts the record from zero, verbatim, so it cannot be lost the same way twice)

SOURCE S-2 request, 19-point hardware spec, 2026-07-18. Read against docs/corporate/LOT\_ROBOTICS\_COSMO.md, docs/corporate/LOT-CUBIQ-VISION.md (Quantum Cube precedent), docs/technical/LOT-NODE-0-RIG-SPEC.md (format precedent), docs/technical/OS\_API.md and live source (POST /api/logs – the real endpoint this device calls).

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## 00 COMPRESSED SUMMARY (session-compression pass, per spec item 8)

ONE SENTENCE: A palm-sized, two-piece stainless steel companion computer that shows one LOT-pushed notification at a time on a round display, watches the room through a small camera and an environmental sensor, and has one button – COPY – that writes a signal to the owner's Log tab on lot-systems.com.

SHAPE: 2-part stainless body, 40mm x 40mm x 5mm flat silver square.  
FRONT: polished stainless steel – no seams, no ports, the calm face.  
BACK: camera + round display + COPY button – the working face.  
POWER: Qi wireless charging, no charge port breaking the shell.  
BRAIN: ESP32-S3 (Wi-Fi, camera ISP, enough RAM for a UI + TLS).  
SENSES: camera (OV2640/OV5640) + BME280/BME680 weather/env sensor.  
LINK: LOT API connector – pulls notifications, POSTs to /api/logs.  
MAKE: PCBWay for PCB + SMT assembly, 100-unit first run.  
PAPER: 3 documents, kept separate on purpose (spec item 11) – this file (hardware/BOM/roadmap), LOT-COMPUTER-FIRMWARE.md, LOT-COMPUTER-SOFTWARE-CONNECTOR.md – plus a generated PDF manual (spec item 7).

RETAINED TOKENS FOR FUTURE SESSIONS: LOT-COMPUTER (product), COPY-SIGNAL (button → /api/logs event), TERMINAL-FACE (display side), MIRROR-FACE (polished blank side). See docs/assembly/LOT-LEXICON.md.

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## 01 PRODUCT THESIS

The LOT Computer is the physical terminal for a LOT profile. It does not run the app – the app already runs on the phone and the web. It does one thing the phone cannot: sit on a desk, ambient and undistracting, and surface exactly one signal from lot-systems.com at a time – a pushed notification like "Coffee time!" – the way a pager surfaces exactly one page. No feed. No scroll. No app icons. A cube-shaped moment, made object, consistent with the CUBIQ® thesis in docs/corporate/LOT-CUBIQ-VISION.md ("physical products – sci-fi hygiene, cleanse, real-world cubic").

It is the entry-level physical product in the COSMO® hardware line, sized and priced to precede the COSMO® robotics unit (docs/corporate/LOT\_ROBOTICS\_COSMO.md, Phase 3, 2028-2029, \$2,500-\$5,000/unit) by roughly two years. The LOT Computer is the low-cost proof that a LOT profile can live outside a browser tab before the company asks anyone to trust it inside a robot.

## 02 FORM FACTOR (spec items 3, 4, 17, 18)

| FRONT – "MIRROR-FACE"                                                                                                                         | BACK – "TERMINAL-FACE"                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Polished stainless steel, mirror (#8) finish. No seam, no port, no branding beyond a laser-etched LOT® mark, filled and polished flush.       | Bead-blasted stainless steel, matte. Cutouts for: |
| Functions as a physical "off" state – face-down, the device shows nothing and reads nothing. Face-up, TERMINAL-FACE is presented to the room. | - camera window (sapphire or Gorilla Glass lens)  |
|                                                                                                                                               | - round display window (same glass)               |
|                                                                                                                                               | - COPY button (stainless, flush, tactile click)   |

BODY: 2-part stainless steel shell, laser-welded or press-fit at the equator seam, gasket-sealed (IP54 target – desk object, not a dive watch).

FOOTPRINT: 40mm x 40mm (4cm x 4cm), flat square, rounded 3mm corners.

HEIGHT: 5mm – deliberately thin. This is the hard engineering constraint of the whole build (see 03, stack budget).

WEIGHT: target 35-45g (stainless is dense; thinness keeps it light).

FINISH: MIRROR-FACE polished #8, TERMINAL-FACE bead-blasted #4 – same two-finish language as LOT-CUBIQ-VISION's "sci-fi hygiene, cleanse" aesthetic.

### ### 02.1 THE 5mm PROBLEM

A 5mm total height is the single hardest constraint in this spec – thinner than an iPhone (7.8mm) and close to a credit card stack (0.76mm x 6 = 4.6mm for reference). It rules out a coin-cell-class battery with any real capacity, rules out a full-size camera module, and forces a rigid-flex or ultra-thin rigid PCB. The roadmap in section 06 treats 5mm as a Phase 2 target, not a Phase 1 guarantee – Phase 1 (proto) is allowed to land at 6-8mm while the stack is proven, then gets thinned for the 100-unit run.

## 03 COMPONENT BILL OF MATERIALS

Prices are current street quotes, July 2026, single-unit vs. estimated 100-unit-run landed cost (component-only, before PCBWay assembly + CNC enclosure – those are priced separately in section 04). Links are real supplier/manufacturer pages, not placeholders – verify current stock and exact SKU before ordering; component markets move week to week the same way NODE-0's GPU pricing does (docs/technical/LOT-NODE-0-RIG-SPEC.md).

| PART                            | SPEC / SKU CLASS                                                                                                                                                                                                                                                                                                                                                                                                                                           | UNIT x1 | UNIT x100 |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------|
| MCU + CAMERA (5)<br>spec item 6 | ESP32-S3 module w/ OV2640 2MP, integrated Wi-Fi/BLE, DVP camera interface, enough PSRAM for a framebuffer + TLS stack.<br>Ref (dev-kit form, for prototyping):<br>Meshnology ESP32-S3 OV2640 board<br><a href="https://www.amazon.com/dp/B0GLX9XCP4">https://www.amazon.com/dp/B0GLX9XCP4</a><br>FORIOT ESP32-S3-CAM N16R8<br><a href="https://www.amazon.com/dp/B0F4DKTBR9">https://www.amazon.com/dp/B0F4DKTBR9</a><br>Production form: bare WROOM-class | \$5-8   | \$2.50-4  |

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |          |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------|
|                                                  | module + discrete OV2640/OV5640 on the custom PCB, not the dev board.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |          |
| DISPLAY<br>spec item "screen"<br>(18)            | 1.28" round TFT, 240x240, GC9A01 driver, SPI. Round face matches the TERMINAL-FACE window; renders one pushed notification at a time, large type, high contrast, no chrome. Waveshare 1.28" round LCD module <a href="https://www.waveshare.com/1.28inch-lcd-module.htm">https://www.waveshare.com/1.28inch-lcd-module.htm</a> Makerfabs GC9A01 round module <a href="https://www.makerfabs.com/gc9a01-1-28-inch-round-lcd-module.html">https://www.makerfabs.com/gc9a01-1-28-inch-round-lcd-module.html</a> NOTE: candidate module is TFT, not e-ink – chosen for instant refresh on a pushed notification. An e-ink variant is a Phase 3 power-savings option (see 06) once refresh latency is proven acceptable.<br>Unit: \$8-15                                                      | \$8-15  | \$4-7    |
| WEATHER / ENV SENSOR<br>spec items 14, 15        | BME280 (temp/humidity/pressure) or BME680 (adds VOC/air-quality – the "AI-grade off-the-shelf sensor" tier). Bosch Sensortec, industry-standard, I2C, small enough for the 5mm stack. Adafruit BME280 STEMMMA QT <a href="https://www.adafruit.com/product/2652">https://www.adafruit.com/product/2652</a> Adafruit BME680 STEMMMA QT <a href="https://www.adafruit.com/product/3660">https://www.adafruit.com/product/3660</a> Bosch BME280 product page <a href="https://www.bosch-sensortec.com/en/products/environmental-sensors/humidity-sensors-bme280">https://www.bosch-sensortec.com/en/products/environmental-sensors/humidity-sensors-bme280</a> Bare-die/module pricing at volume runs well under the breakout-board price.<br>Unit: \$9-19 (breakout) / ~\$3-5 (bare, x100) |         |          |
| BUTTON<br>spec item 16                           | Single stainless tactile switch, IP54-rated, flush-mount, laser-etched "COPY". Any qualified SMT tactile switch behind a stainless cap; supplier selected at CM (contract manufacturer) stage – not BOM-critical, commodity part.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <\$1    | <\$0.40  |
| WIRELESS CHARGE<br>RECEIVER<br>spec items 12, 19 | Qi receiver module, small coil (~10-15mm diameter to fit the 40mm footprint), 5V/1A out, feeds a buck/charge-management stage. Adafruit Universal Qi Receiver <a href="https://www.adafruit.com/product/1901">https://www.adafruit.com/product/1901</a> Small-coil Qi receiver reference <a href="https://www.wirelesschargingcoil.com/small-wireless-charging-coil/">https://www.wirelesschargingcoil.com/small-wireless-charging-coil/</a>                                                                                                                                                                                                                                                                                                                                             | \$3-8   | \$1.50-3 |
| BATTERY                                          | Ultra-thin LiPo pouch cell, 40-80mAh, sized to the 40x40x5mm envelope (a full 500mAh pouch will NOT fit at 5mm total height – see 02.1; Phase 1 proto uses a larger cell and accepts a thicker enclosure).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | \$2-5   | \$1-2    |
| CHARGE MANAGEMENT IC<br>spec item 12             | TP4057 (500mA linear Li-ion charger, SOT-23-6) between the Qi receiver and the battery. TP4057 datasheet <a href="https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/5010/TP4057.pdf">https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/5010/TP4057.pdf</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <\$0.15 | <\$0.11  |
| PCB (bare)<br>spec item 1                        | Custom rigid or rigid-flex, 2-4 layer, sized to the 40x40mm footprint, fabricated by PCBWay. PCBWay PCB quote <a href="https://www.pcbway.com/HighQualityOrderOnline.aspx">https://www.pcbway.com/HighQualityOrderOnline.aspx</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | \$2-6   | \$0.80-2 |

[illegible]

| STAGE | ACTION                                                                                                                                                                                                                                    | VENDOR / TOOL                                                                                                                            |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Panelize the 40x40mm board – target 4-up or 6-up per panel to clear the 50x100mm assembly minimum.                                                                                                                                        | PCBWay quote tool<br><a href="https://www.pcbway.com/HighQualityOrderOnline.aspx">https://www.pcbway.com/HighQualityOrderOnline.aspx</a> |
| 2     | Bare PCB fab, 2-4 layer, ENIG finish (small pads, fine-pitch ESP32-S3 module).                                                                                                                                                            | PCBWay                                                                                                                                   |
| 3     | Turnkey SMT assembly – PCBWay sources BOM parts, places, reflows, AOI + functional test per unit.                                                                                                                                         | PCBWay SMT quote<br><a href="https://www.pcbway.com/quotesmt.aspx">https://www.pcbway.com/quotesmt.aspx</a>                              |
| 4     | X-ray inspection on the ESP32-S3 module footprint (BGA/QFN-class package, hidden joints).                                                                                                                                                 | PCBWay                                                                                                                                   |
| 5     | V-score or tab-route depanelization, 100 boards.                                                                                                                                                                                          | PCBWay / CM                                                                                                                              |
| 6     | Enclosure run in parallel – CNC or deep-draw stainless, 100 units, both faces finished, laser-etched.                                                                                                                                     | Xometry (or<br>PCBWay's own<br>CNC service –<br>get both<br>quotes)                                                                      |
| 7     | Final assembly: PCB into shell, lens/window bonded, gasket seated, wireless-charge coil aligned, unit function test (charge, camera frame, display render, button → /api/logs round-trip against a staging LOT API endpoint) before ship. | Contract<br>manufacturer<br>(CM) – PCBWay<br>offers this as<br>an add-on; a<br>dedicated CM<br>is the Phase<br>3 default.                |

FIRST RUN SIZE: 100 units, per spec item 13. This is a pilot-production quantity – large enough to amortize PCB panel tooling and get a real CNC quote, small enough that a design flaw costs a batch, not a warehouse.

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## 05 DOCUMENTATION SET (spec items 7, 9, 10, 11)

Per spec item 11, firmware and software documentation are kept as SEPARATE files – not sections of this one – so each can version independently as the MCU-side and backend-side implementations diverge.

| DOCUMENT<br>■■■■■■■■■                     | COVERS<br>■■■■■■■                                                                                                                   |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| LOT-COMPUTER-HARDWARE-SPEC.md (this file) | Plan, BOM, roadmap, PCBWay plan, form factor.                                                                                       |
| LOT-COMPUTER-FIRMWARE.md                  | MCU firmware: boot, camera driver, display render loop, button/COPY signal handling, wireless-charge power management, OTA.         |
| LOT-COMPUTER-SOFTWARE-CONNECTOR.md        | LOT API connector: pairing/auth, notification pull, POST /api/logs schema, retry/offline behavior.                                  |
| LOT-COMPUTER-HARDWARE-SPEC.pdf            | Generated PDF manual (spec item 7) – this document, print-ready, filed alongside the badge codex PDFs precedent (docs/badges/pdf/). |

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## 06 ROADMAP

| PHASE<br>■■■■■ | NAME<br>■■■■■     | TARGET<br>■■■■■■■ | KEY WORK<br>■■■■■■■■■                                                                                                                                                                                                                                                                       |
|----------------|-------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0              | Paper             | 2026 Q3 (now)     | This spec. BOM priced. PCBWay + CNC quotes requested. Firmware + software connector docs authored (this session).                                                                                                                                                                           |
| 1              | Bench Prototype   | 2026 Q4           | ESP32-S3-CAM dev board + breakout GC9A01 display + BME280 breakout, wired dead-bug on a bench. Thickness constraint relaxed (accepts >5mm). Proves: camera capture, round-display render of a pushed "Coffee time!" notification, COPY button -& POST /api/logs round-trip against staging. |
| 2              | Custom PCB Rev A  | 2027 Q1           | Panelized custom board, PCBWay fab + assembly, 10-25 units. First real attempt at the 40x40x5mm envelope. Wireless charging integrated.                                                                                                                                                     |
| 3              | Enclosure + Rev B | 2027 Q2           | Stainless shell (Xometry or PCBWay CNC) fitted to Rev A/B board. IP54 seal validated. Battery sized to final cavity.                                                                                                                                                                        |
| 4              | 100-Unit Run      | 2027 Q3           | Section 04 pipeline, full 100 units, CM final assembly + function test.                                                                                                                                                                                                                     |
| 5              | Fleet Pairing     | 2027 Q3-Q4        | Software connector live                                                                                                                                                                                                                                                                     |

against lot-systems.com  
production, per-unit  
pairing flow, OTA  
firmware channel open.

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**07 OPEN RISKS (recorded honestly, not smoothed over)**

| RISK                                                                                                   | MITIGATION                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ■■■■■                                                                                                  | ■■■■■■■■■■                                                                                                                                                                                                       |
| 5mm height may not fit camera + coil + battery + display stack simultaneously                          | Phase 1/2 explicitly allowed to exceed 5mm; thinness is a Phase 3 target, not a Phase 1 gate.                                                                                                                    |
| 40x40mm PCB is below PCBWay's stated 50x100mm assembly minimum                                         | Panelize from the first fab run (section 04, stage 1).                                                                                                                                                           |
| Round display + camera window in a 5mm stainless shell = two sealed optical cutouts in a metal shell   | Sapphire/Gorilla-glass bonding is a real yield risk at 100 units; budget for a higher scrap rate on the enclosure line than the PCB line.                                                                        |
| No physical charge port (Qi-only) means a bricked unit cannot be recovered by wired debug in the field | Firmware must expose a clear low-battery / boot-fail signal on the display before it goes dark; OTA and pairing flow must be resilient to a mid-update power loss (see LOT-COMPUTER-FIRMWARE.md 06 OTA section). |

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